SIGMIDI Foot Controller v0.1 — Hardware and wiring

Target: Arduino Nano ESP32 (ABX00083, ESP32-S3), using its internal ADC, and Waveshare 24382 1.69-inch ST7789V2 color LCD, used in landscape at 280×240. The harness requires bench validation; verify against the actual board revision before assembly. All logic and analog circuits use 3.3 V and common ground. Only one footswitch needs to be detected at a time.

Alternate power input

Use a **9 V DC center-negative 2.1 × 5.5 mm barrel supply**. The positive sleeve feeds **Nano VIN** and Pololu VIN; the negative centre feeds common GND. The Pololu S13V10F5 (4083 / Kamami 1178565) supplies regulated **5 V to the optional LEDs**, not Nano VIN. Nano VIN requires 6–21 V. USB can power the Nano independently; do not tie external 5 V to USB VBUS. See power wiring. The Nano has no LiPo charging circuit.

Wi-Fi configuration access

Use the Nano ESP32 onboard Wi-Fi radio for the configuration editor. No additional module, wiring, or GPIO is required. Wi-Fi is enabled explicitly for setup and disabled when returning to performance mode; see the firmware specification for connection and save behavior.

GPIO assignments

The firmware uses **ESP32 GPIO numbers**, independently of Arduino IDE pin-numbering preferences. The Nano's board labels and corresponding GPIOs are:

Function	Nano label	ESP32-S3 GPIO	Connection
Switch ladder	A0	1 / ADC1_CH0	Ladder sense
Pedal 1	A1	2 / ADC1_CH1	Direct TIP
Pedal 2	A3	4 / ADC1_CH3	Direct TIP
Pedal 1 bias	D2	5	10 kΩ to pedal 1 TIP
Pedal 2 bias	D3	6	10 kΩ to pedal 2 TIP
Encoder A	D4	7	Contact A
Encoder B	D5	8	Contact B
Encoder click	D6	9	Contact to GND
Optional LED DIN	D7	10	Direct to first Pololu 2535 DIN
LCD CLK	D8	17	SPI clock (routed through GPIO matrix)
LCD DC	D9	18	Data/command
LCD CS	D10	21	Active-low select
LCD DIN	D11	38	SPI MOSI
LCD RST	D12	47	Active-low reset
UART1 MIDI OUT TX	D0 / RX0 pad	44	UART1 TX to 74HCT125 TRS output stage
USB VBUS sense	A4	11	100 kΩ / 150 kΩ divider below

Function	Nano label	ESP32-S3 GPIO	Connection
LCD VCC and BL	3V3	—	Always-on backlight
All grounds	GND	—	Common return

UART1 is explicitly mapped to GPIO44 (the Nano labels this header pad RX0), leaving D1/GPIO43 UART0 TX free of ROM boot text. D1 is intentionally unwired; D0's normal RX function is not used. UART1 MIDI OUT runs at 31,250 baud, 8-N-1 and does not alter the USB-C device or flashing interface.

The TRS MIDI OUT stage runs from `MIDI_5V`, a Schottky diode-OR of USB VBUS and converter +5 V. The separate diodes prevent source back-feed.

The display's eight-pin connector stays in physical order **BL, RST, D/C, CS, SCL/CLK, SDA/DIN, GND, VCC**. Connect to the signals above. The Nano's D13/default SPI-clock pin is deliberately unused because it also drives the onboard LED. GPIO19/20 are exclusively USB D−/D+ inside the USB-C connection. Avoid GPIO0/3/45/46 strapping pins and leave unused Nano pins disconnected. The three analog inputs all use ADC1, including during Wi-Fi operation. Pin constants and uniqueness checks are in `src/board.hpp`.

USB VBUS detection

Install these **two resistors even if you initially use only BLE**:

```
Nano VBUS (USB 5 V) --- R50 100 kΩ ---+--- A4 / GPIO11
                                      |
                                   R51 150 kΩ
                                      |
                                     GND
```

Use 1% resistors. The divider gives about 3.0 V with a 5 V USB supply and discharges the sense input when unplugged. Never connect 5 V directly to GPIO11. This monitors host VBUS while the controller remains powered through VIN; it does not supply power or join external 5 V to USB. Native USB MIDI uses the existing USB-C connector and needs no external USB adapter. The TRS MIDI OUT stage takes its 5 V rail from a Schottky diode OR of USB VBUS and the converter’s +5 V output; the diodes prevent either source from feeding the other.

LCD VCC → 3V3 and GND → GND; use the factory four-wire SPI configuration. No MISO signal is needed. Connect BL to 3V3 for always-on backlight; no additional GPIO is allocated. The display is not touch-sensitive. The module uses its supplied GH1.25 eight-pin connector; follow the signal names on the module, rather than wire colors. Landscape orientation uses a 280×240 drawing area, MADCTL 0x70 (RGB), RGB565 pixels, inversion on, and column/row offsets (20, 0): controller columns 20–299 and rows 0–239. The driver runs SPI at 10 MHz and applies the landscape mapping after vendor initialization. The saved Enclosure variant selects Right (0° screen, Right switch order) or Left (180° screen, mirrored switch order) through software stripe/window mapping; controller register settings remain the same. New configurations default to the preferred Left enclosure with the screen and ribbon on the left. For an existing saved setup choose Left, using Wi-Fi Settings → Enclosure variant or LCD EDIT CONFIG → ENCLOSURE. Verify physical viewing direction and red/blue color order on the actual module. The pin assignment reserves native USB and boot straps.

Buffered TRS MIDI OUT

The one-way output uses UART1 TX on the Nano D0/RX0 pad at **GPIO44**, at 31,250 baud, 8 data bits, no parity, 1 stop bit. UART0 D1/TX0 GPIO43 stays unconnected to avoid boot-ROM output. The Nano source pin drives the joined inputs of 74HCT125 gates 1 and 2 (pins 2 and 5); their active-low enables (pins 1 and 4) are grounded, the supply is `MIDI_5V`, and a 100 nF ceramic bypass is placed at the IC supply pins. Output 1Y (pin 3) drives J3 and output 2Y (pin 6) drives J4, so each socket has its own buffer. Tie the unused gate inputs (pins 9, 12) and enables (pins 10, 13) to GND; leave outputs 8 and 11 open. A 10 kΩ resistor from the buffer inputs to 3V3 holds them at MIDI idle-high during reset. Do not connect a MIDI loop directly to the Nano GPIO.

Use a diode-OR supply: a Schottky diode from converter +5V and a second from USB VBUS feed `MIDI_5V` (anodes to sources, cathodes joined). This powers the interface in USB-only and 9 V/converter operation while preventing source back-feed. Each 3.5 mm TRS socket uses the MIDI TRS Type A mapping and its own current

loop: `MIDI_5V` → 220 Ω → ring (DIN pin 4) → receiver optocoupler → tip (DIN pin 5) → 220 Ω → its buffer output. J3 uses R53/R54 and 1Y; J4 uses R55/R56 and 2Y. Each sleeve (DIN pin 2, shield) connects to GND so the cable shield is grounded at the sending end. Identify the tip, ring and sleeve lugs on the selected socket before wiring. Use Type A cables or adapters for 5-pin DIN receivers. The USB-C connection remains native USB MIDI and ROM flashing; UART1 is a separate physical MIDI OUT.

The TRS outputs have not been electrically bench-qualified. Before attaching equipment, check diode direction, 5 V rail isolation, tip/ring wiring and output current-loop polarity on both sockets with a current-limited supply and MIDI test receiver.

Carrier perfboard

One 80.5 x 20 x 1.5 mm perfboard carries every resistor and capacitor that connects directly to the Nano (R7/R8/C5 and the first ladder segment R9, R20/R21, R30/R31, R50/R51, C1), together with the complete TRS MIDI OUT stage (R52, D1/D2, U4 in a DIP-14 socket, C6, R53-R56). The board has a 6 x 28 hole field at 2.5 mm pitch, four 3 mm mounting holes on a 76 x 16 mm rectangle and four 2 x 3.5 mm pads at each end, between its mounting holes; the parts use columns 0-25. It sits at base level in the rear, between the Nano and the enclosure side, under the dual TRS socket. Off-board wires enter at labelled holes. Assembly guide step 7 gives the placement, trails, insulated links and wire-entry table; `scripts/perfboard_layout.py` checks the layout against the netlist.

Rotary encoder

Add a passive incremental quadrature encoder with 20 detents per revolution and an independent momentary push switch. Confirm the selected part's pulses-per-revolution and detent phase relationship: 20 detents does not necessarily mean 20 complete quadrature cycles.

- Connect encoder common C to GND, A directly to D4 / GPIO7, and B directly to D5 / GPIO8.
- Connect one push-switch terminal directly to D6 / GPIO9 and the other to GND.
- No external resistors or capacitors are required. Firmware enables the GPIO internal pull-ups for A, B and click, decodes quadrature transitions and debounces the click in software. Keep encoder wires short.
- Use a bare contact encoder or a module explicitly compatible with 3.3 V; verify terminal identification from the selected encoder datasheet.

Validate one menu step per physical detent in both directions, including slow turns, quick turns, reversal, and rotation while clicking. The encoder operates independently of the footswitch ladder.

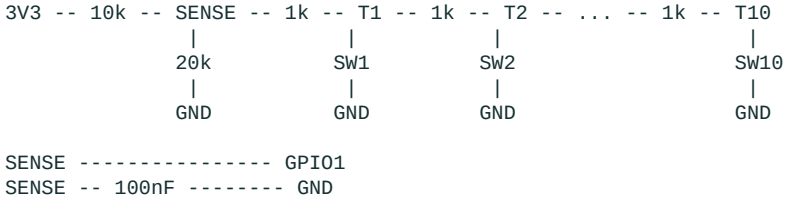
Modular series-tap resistor ladder

Use a four-switch base plus 0–3 two-switch extensions for 4/6/8/10 total. Extensions go to the right in the preferred Left enclosure and to the left in Right. Keep the 10 kΩ pull-up, 20 kΩ idle resistor and 100 nF capacitor in the electronics base. Each extension adds two 1 kΩ series sections and carries TAIL and GND onward.

Tap identity is fixed: A=T2, B=T1, 1=T4, 2=T3; extensions add C/3=T5/T6, D/4=T7/T8 and E/5=T9/T10. Select Modular wiring and the installed count. Wiring, resistor values, labels and voltage targets.

The schematic below shows electrical tap order, independently of physical row order. Each SWn connects tap Tn to ground.

Use 1% metal-film resistors (0.1% optional) and normally open switches. Build this series-tap topology:



There are ten 1 kΩ series sections: SENSE–T1 through T9–T10. No extra connection terminates T10. Each switch connects only its tap to ground. The permanent 20 kΩ resistor keeps the idle input below the ADC measurement ceiling. Place the 100 nF capacitor close to GPIO1. Start with 10 ms debounce and check the settling and noise on the assembled harness.

With switch n closed, the lower resistance is `(n × 1k) || 20k`. Ideal voltages at 3.300 V are:

Switch	Series resistance to grounded tap	Sense voltage
1	1 kΩ	0.287 V
2	2 kΩ	0.508 V
3	3 kΩ	0.683 V
4	4 kΩ	0.825 V
5	5 kΩ	0.943 V
6	6 kΩ	1.042 V
7	7 kΩ	1.127 V
8	8 kΩ	1.200 V
9	9 kΩ	1.264 V
10	10 kΩ	1.320 V
None	Open taps	2.200 V

These are calculated targets, not measured ADC thresholds. Capture each switch and idle during calibration; use narrow acceptance windows with guard bands, not a nearest-value classifier across the entire ADC range. The smallest nominal separation is about 56 mV. Verify separation with resistor tolerances, contact resistance, supply variation, BLE traffic, and LCD activity. Require measured noise/drift plus component uncertainty to remain comfortably within half the adjacent separation; otherwise revise the ladder before enclosure assembly.

Operation is one footswitch at a time. Simultaneous presses are outside the requirements, so no combination decoding is needed. Firmware waits for stable idle before accepting the next press.

Two selectable pedal inputs

Each input supports a two-wire variable resistor or a three-wire potentiometer, independently selected. TRS wiring: tip = signal, ring = reference, sleeve = ground. Verify actual pedal polarity; provide a tip/ring swap adapter where needed. Two-wire pedals connect signal to ground through their variable resistance. A TS plug may short ring to sleeve, so the reference output must tolerate that condition.

Each input needs **only two resistors inside the controller**, with no op-amp, MOSFET, clamp circuit, divider or external filter capacitor:

Connection	Pedal 1	Pedal 2
TIP directly to ADC	A1 / GPIO2	A3 / GPIO4
RING reference	3V3 through 5.1 kΩ	3V3 through 5.1 kΩ
SLEEVE	GND	GND
10 kΩ from bias pin to TIP	D2 / GPIO5	D3 / GPIO6

Use 1% resistors. A 5.1 kΩ ring resistor limits a ring-to-ground short to about 0.65 mA at 3.3 V. It also limits a 10 kΩ potentiometer's nominal full-travel voltage to `3300 × 10000 / (5100 + 10000) = 2185 mV`; lower-resistance pots have a lower full-travel voltage. TIP is measured directly without a gain correction. A 10

kΩ pot's maximum source impedance is approximately 3.78 kΩ; validate acquisition settling and noise on the actual harness.

For an enabled **two-wire variable resistor**, firmware drives the bias pin HIGH through its 10 kΩ resistor. The external resistance connects TIP to SLEEVE. Use `Rpedal = 10000 × Vtip / (3300 - Vtip)` with nominal millivolts, then normalize against the configured maximum resistance (positive, at most 10 kΩ). A 10 kΩ resistor gives nominal 1650 mV. The GPIO HIGH voltage and resistor tolerance affect resistance accuracy. Open circuit approaches the 3.3 V supply; firmware guards the denominator near that value. A TS plug may ground RING safely through the 5.1 kΩ reference resistor.

For a **three-wire potentiometer**, connect the outer legs to RING and SLEEVE and the wiper to TIP. Firmware leaves the bias GPIO at high impedance with both internal pulls disabled. Capture the actual heel and toe voltages before use; these endpoints set the position scale independently of pot resistance. A floating wiper is not a dependable unplug detector. The external pedal contains only the resistor or potentiometer, without any powered electronics.

Boot and disabled-input states also leave the bias GPIO at high impedance. The firmware preloads HIGH before enabling its output and never drives the bias LOW. Change mode with the pedal unplugged and repeat endpoint calibration after rewiring. ADC voltages are used directly without a gain multiplier. Internal weak pull-ups are not used as precision resistance references.

ADC and prototype validation

Use calibrated ESP32-S3 ADC1 readings at 12 dB attenuation, 12-bit resolution and the ESP-IDF curve-fitting scheme, with one calibration handle per input. Connected pedals and idle ladder remain within the nominal 0–2.2 V design span. Calibrate actual switch centres and pedal endpoints on the installed Nano. Absolute ADC error, acquisition settling and noise still require bench checks. Ladder scans run about every 1 ms; pedal sampling starts at 200 samples/s. Connect only unpowered passive pedals. The two-resistor pedal interface adds no external ESD protection.

Before building the harness, verify idle and all N installed ladder states, individual press/release detection, and all four combinations of pedal modes. Test with BLE and LCD active, USB and 9 V power operation, and the longest intended pedal cables. Check the Nano and LED supply budgets.

Optional switch LEDs

The optional chain uses one Pololu 2535 5 mm LED per switch, initially paired in switch-number order. Use **5 V, RGB byte order and the 2535 (800 kbit/s) profile**. The side-view lead order is DIN / +5 V / GND / DOUT, from the side opposite the flat toward the flat; follow the orientation diagram in the LED guide. D7 / GPIO10 is reserved for its hardware-timed data output. The application exposes USB MIDI rather than a serial console. Connect GPIO10 directly to the first DIN, each DOUT to the next DIN, and every LED to the regulated 5 V supply and common ground. The chain contains only LEDs and wires. Ten LEDs draw about 0.5 A at full white; a separate LED supply should allow at least 1 A plus controller headroom. When using the optional Pololu 4083 source, its 1 A rating is the total controller-plus-LED budget; follow the limitations in the power guide. Never power this part above 5 V. Chain output is disabled by default, with 64/255 brightness and white/yellow press/release colors. Pololu LEDs power up blue until a data command; firmware clears them at initialization and refreshes the chain every 250 ms. See exact lead orientation, wiring and configuration.

Sources

- Arduino Nano ESP32 documentation and physical pinout.
- Espressif USB device stack: native USB and self-powered VBUS detection.
- Waveshare 24382 module wiki: ST7789V2, interface configuration and physical connector.

The electrical diagrams and KiCad schematic use this Nano GPIO/power map and preserve physical pin order. The enclosure uses the direct Nano base mounts.